

PROBLEM STATEMENT

ComicCraft – AI Comic Story Creator using Gemini Models

1. Background

Comic creation is a creative process that combines storytelling, character development, scene planning, dialogue writing, visual design, and illustration. A creator must convert a simple story idea into a sequence of meaningful scenes and comic panels while maintaining a clear flow from beginning to end. For students, beginners, and users who are interested in creating comics, completing all these activities manually can require considerable time and creative effort.

With the growth of generative AI, there is an opportunity to support the comic creation process through intelligent story and visual generation. ComicCraft is proposed as an AI-based web application that helps users transform a basic story idea into structured comic content using Gemini models.

2. Existing Problem

Traditional comic creation requires the user to perform several separate tasks. The user must first develop the story, identify characters, decide what happens in each scene, write suitable dialogues, and then create or arrange illustrations. These activities are often performed using different tools or through manual work. As a result, the overall process can become lengthy and difficult for users who do not have strong writing, drawing, or visual-design skills.

Even when a creator has a good story idea, converting that idea into a complete comic may be challenging. Maintaining continuity between scenes, keeping characters relevant to the story, and presenting dialogues together with suitable visuals can require repeated planning and editing.

3. Major Challenges Faced by Users

- **Story Development:** Converting a basic idea into a complete and logically structured story can be difficult.
- **Character Creation:** Users may struggle to define suitable characters and their roles within the story.
- **Scene Planning:** Breaking the story into meaningful scenes and deciding what should happen in each scene requires planning.
- **Dialogue Writing:** Creating dialogues that match the characters and events of each scene can take significant effort.
- **Illustration:** Creating comic visuals requires drawing ability, design knowledge, or additional tools.
- **Time Consumption:** Performing story writing, planning, illustration, and editing separately increases the total creation time.
- **Consistency:** Keeping the story, characters, scenes, dialogues, and illustrations aligned throughout the comic can be challenging.
- **Technical and Creative Skills:** Beginners may not have all the skills required to complete a comic independently.

4. Need for the Proposed System

There is a need for a simple and integrated solution that can assist users throughout the comic creation process. Instead of requiring the user to manually complete every stage, an AI-assisted system can accept a story idea and help generate structured content such as characters, scenes, dialogues, and related

illustrations. This can reduce repetitive work and make comic creation more accessible to beginners and students.

ComicCraft addresses this need by providing a web-based interface connected to a Flask backend and Gemini models. The system is designed to process the user's story idea and support the generation of the major components required for a comic.

5. Detailed Problem Statement

The main problem addressed by ComicCraft is the difficulty of converting a simple comic story idea into a complete, structured, and visually supported comic through manual effort. Users may have creative ideas but face difficulties in writing a complete storyline, developing characters, dividing the story into scenes, preparing dialogues, and creating illustrations. The absence of an integrated and easy-to-use process can make comic creation time-consuming and demanding, particularly for users with limited artistic or writing skills.

Therefore, a system is required that can use AI capabilities to assist users in transforming their initial story concept into organized comic content. The proposed ComicCraft system aims to provide this assistance through Gemini-based generation and a user-friendly web interface, allowing the generated story components and visual content to be presented in a structured manner.

6. Impact of the Problem

- Increased time required to create a complete comic.
- Greater dependence on writing, drawing, and visual-design skills.
- Difficulty for beginners in converting ideas into finished comic content.
- Additional effort required to plan characters, scenes, dialogues, and illustrations separately.
- Difficulty in maintaining continuity and organization across the comic.

7. Project Goal

The goal of ComicCraft is to develop an AI-assisted comic story creation platform that simplifies the process of turning a user's story idea into structured comic content. The system aims to support story generation, character creation, scene generation, dialogue generation, and comic illustration generation through Gemini-based AI capabilities.

8. Expected Improvement

The proposed system is expected to make the comic creation workflow more accessible, organized, and efficient. By bringing AI-assisted story and visual generation into a single web-based application, ComicCraft can reduce the amount of manual effort involved in the early stages of comic creation and help users focus more on their creative ideas. The system is particularly intended to support students, beginners, and aspiring comic creators who want to develop comic stories without needing advanced drawing or writing skills.